# Week 15 Capstone — Dataset Selection Guide

Your team picks its own dataset. This is guidance, not a checklist you need approval on — there's no gate to pass before you start. Use it to avoid picking something that turns out not to work halfway through Week 15.

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## What your data needs to support

Your capstone applies four methods to your data. Before committing to a dataset, check it can support all four:

| Method | What you need |
|---|---|
| **Regression** (Section B) | A numeric **predictor** (X) and a numeric **outcome** (Y) you have a real reason to expect are related |
| **Hypothesis test** (Section C) | A categorical or binary **grouping variable** that splits your rows into two meaningful groups |
| **Monte Carlo simulation** (Section D) | Enough rows and enough understanding of your predictor's natural variability to justify an uncertainty assumption |
| **Bootstrap CI** (Section B4–B5) | Enough rows for resampling to be meaningful — **100+ rows recommended**; more is better |

**The grouping has to be one your data already comes with.** A category that genuinely distinguishes your rows — wet season and dry season, before and after a retrofit, inspected and uninspected, two different sites — is a real grouping, and a date column split at a real event counts. What does not count is cutting a continuous variable at its median to manufacture two halves. That comparison is close to guaranteed to come out "significant" no matter what your data holds, because you built the difference yourself when you chose where to cut; it tells your client nothing, and it is the one move most likely to sink a capstone that is otherwise sound.

If your data has no real grouping in it, pick different data. That is a five-minute decision in Week 13 and a lost weekend in Week 15.

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## Suggested public data sources

These are starting points, not a restricted list — any legitimate public dataset works.

**Transportation:** - ADOT Open Data Portal — Arizona traffic counts, crash data - FHWA / Bureau of Transportation Statistics — national traffic, freight data - City of Tucson open data portal — local traffic counts, transit ridership

**Water resources / hydrology:** - USGS National Water Information System — streamflow, groundwater levels - NOAA Climate Data Online — precipitation, weather records (same source as Weeks 1–2's Tucson data) - Bureau of Reclamation — reservoir levels, water use

**Structural / geotechnical:** - USGS Earthquake Catalog — seismic event data - State DOT bridge inspection databases (many states publish these) — structural condition ratings over time - National Bridge Inventory — condition, load ratings, age

**Energy / buildings:** - EIA (Energy Information Administration) — electricity consumption, rates - 5 ready-made building-energy datasets already in the course files (`Week15_Hospital_Dataset.csv` and similar) — no search required

**General:** - data.gov — federal open data across every domain above - Kaggle Datasets — searchable, often pre-cleaned (verify the source is legitimate/documented before relying on it)

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## A few practical notes

- **Download a sample before committing.** Open the CSV, check the actual column names and row count match what you expect — public data portals sometimes have thin documentation.
- **Messy data is normal.** If your dataset needs cleaning (missing values, inconsistent formatting), that's a legitimate part of real engineering work — just budget time for it.
- **You don't have to pick something novel.** Using one of the 5 legacy building-energy datasets is a completely valid choice if you'd rather focus your time on the analysis than the search.
- **Ask your team "so what?" before finalizing.** If you can't articulate why a client would care about the relationship you're about to model, that's worth reconsidering before you're several hours into the notebook.